

Design Phase

Clean Serve



Supervisor

Submitted by

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1. Use Case Diagram

The use case diagram offers a high-level view of the Clean Serve system by illustrating its main functionalities and the actors who interact with them. It identifies key users such as

Admin, Customer, and Kitchen Staff, and maps their interactions with essential features like Table Reservation, Online Ordering, Billing, Inventory Management, and AI-driven Hygiene Monitoring.

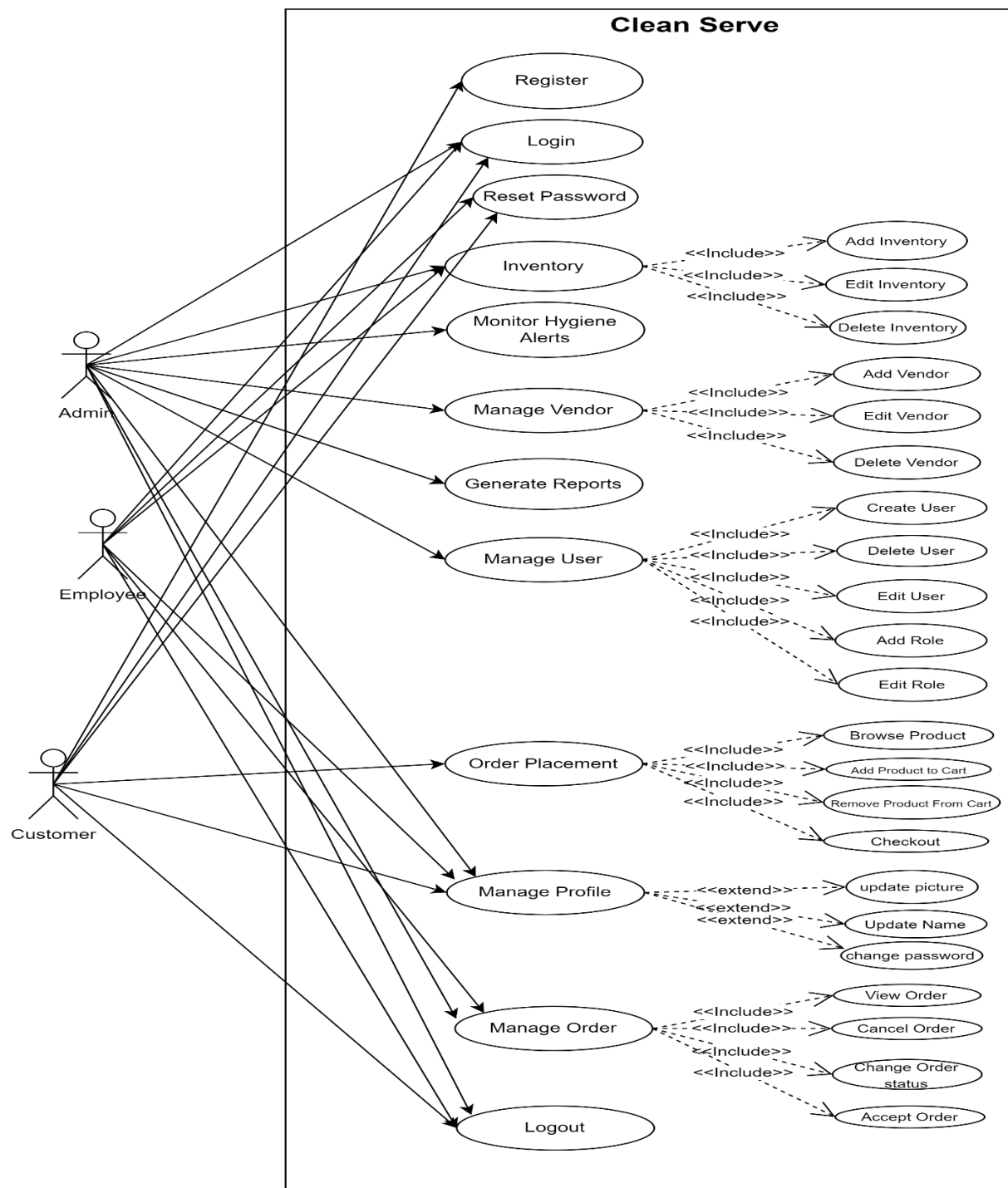


Figure 1: Use Case Diagram – Clean Serve System

2. Sequence Diagram

Sequence diagrams illustrate the flow of messages exchanged between different objects or actors within a specific use case, showing the interaction step-by-step over time. For example, the sequence

diagram for the "Place Order" functionality depicts how a customer initiates an order through the web application, how the order service processes the request, and how the kitchen or admin is notified to prepare the order.

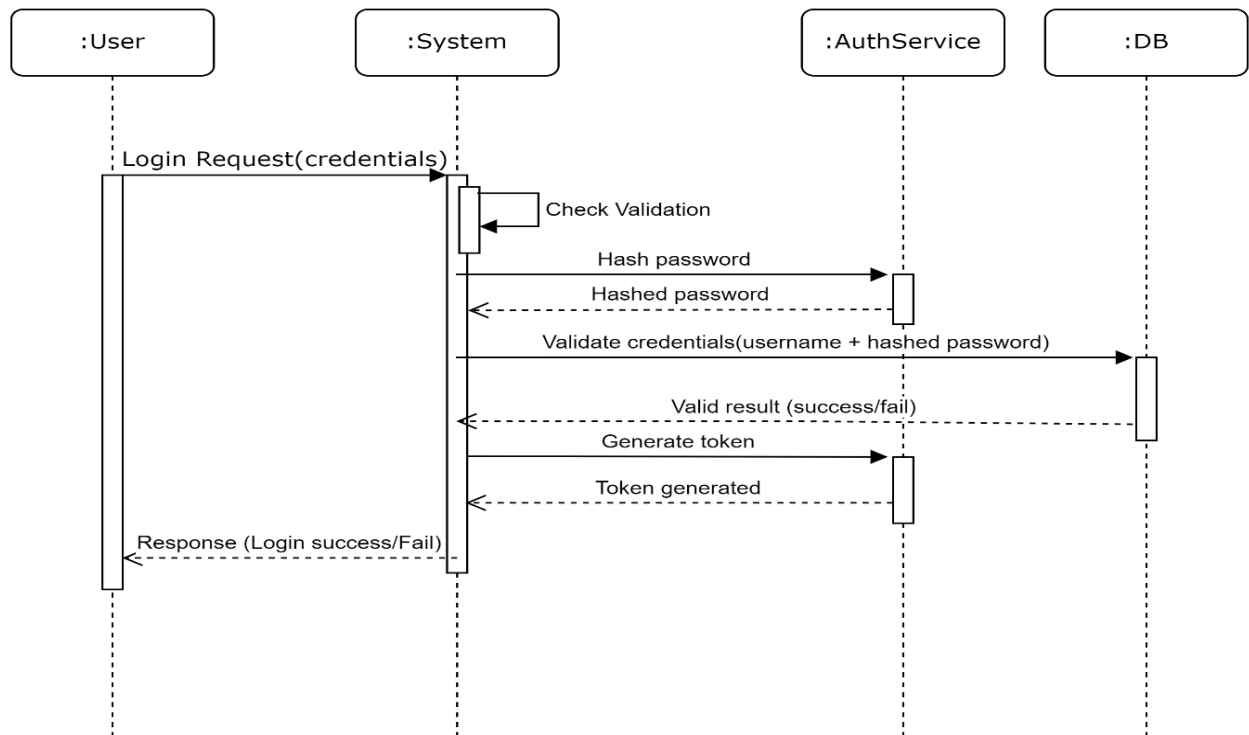


Figure 2: Sequence Diagram for Login

- Represents core functionalities of a job-based platform including login, profile, and job management.
- Uses **<<include>>** for mandatory sub-actions (e.g., add/edit post) and **<<extend>>** for optional features (e.g., update skills).
- Covers full workflow: register → apply/assign job → manage status → complete → review/dispute.
- Separates admin/user roles for actions like user and post management.

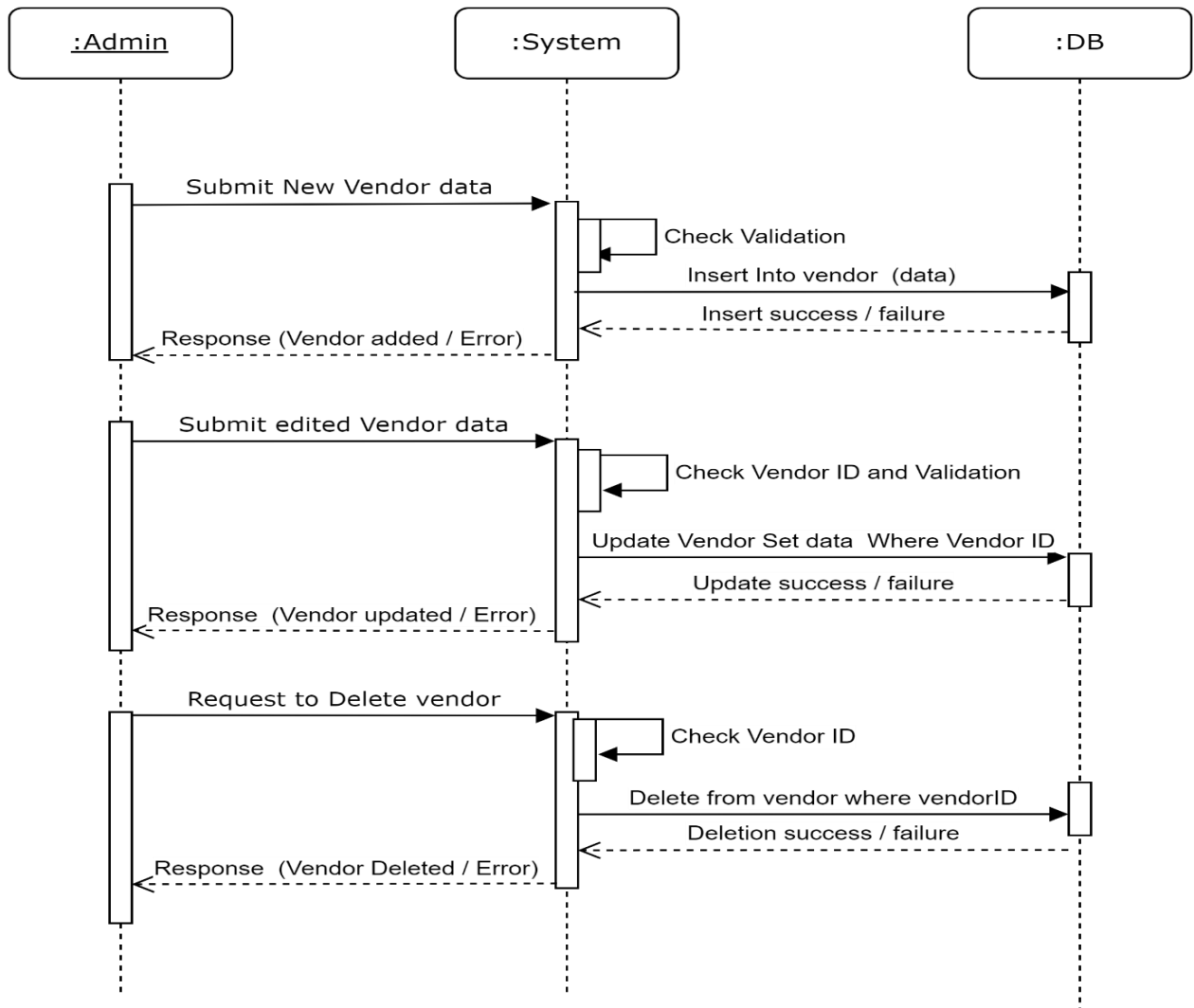


Figure 3: Sequence Diagram for Vendor Management

- **Admin initiates vendor management actions:** Add, Update, or Delete a vendor.
- **System performs validation:** Checks data and/or vendor ID before processing.
- **System interacts with the Database:**
- **Add:** Inserts new vendor data.
- **Update:** Updates existing vendor data using the vendor ID.
- **Delete:** Deletes vendor using the vendor ID.
- **Database returns success or failure status** for each operation.
- **System sends a response back to Admin** indicating the result (success or error).

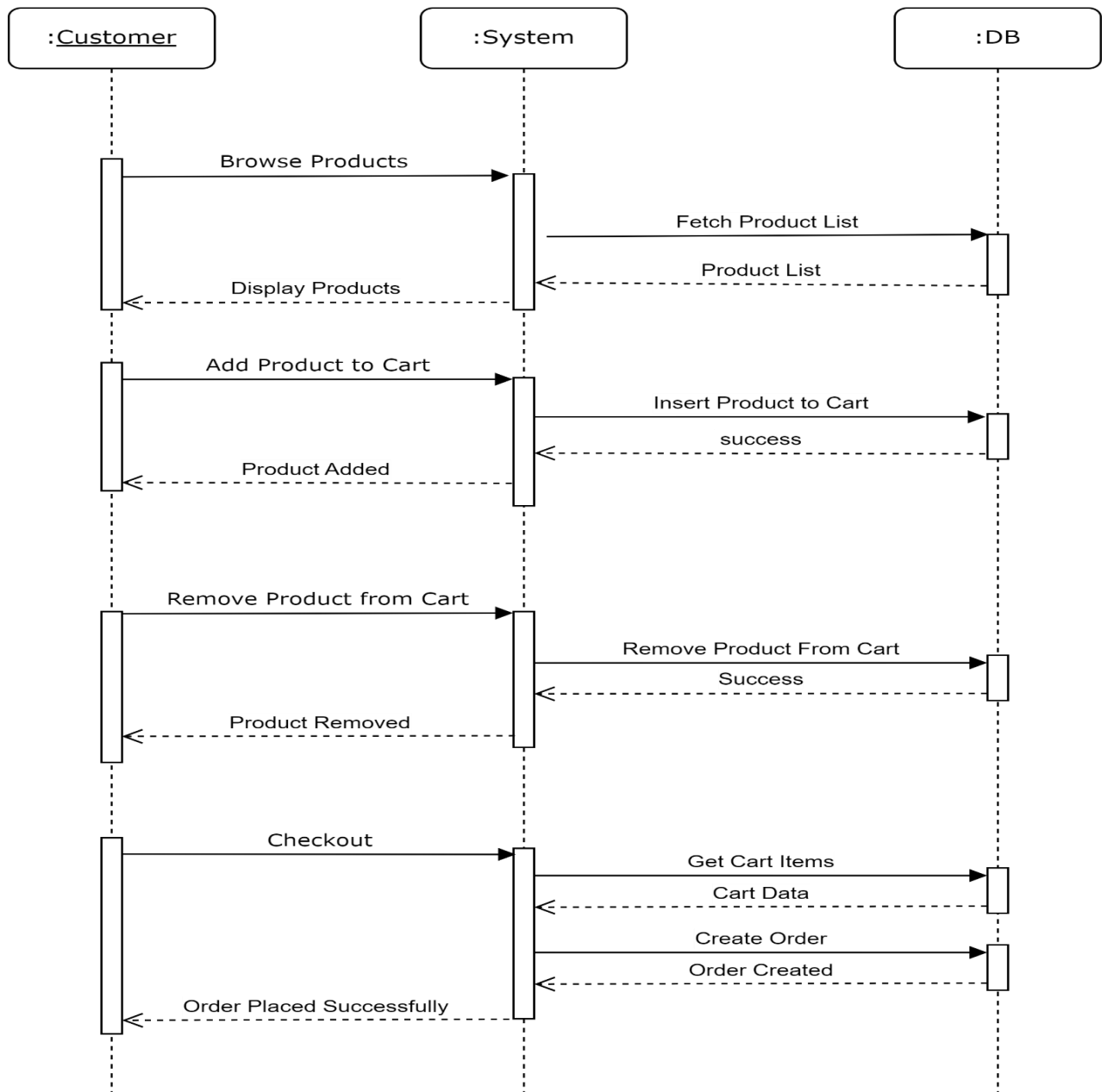


Figure 4: Sequence Diagram for Order Placement

- Admin can **Add, Update, or Delete** a vendor.
- System **validates** input and vendor ID.
- System interacts with **Database** to perform actions.
- Database returns **success or failure**.
- System sends **response** back to Admin.

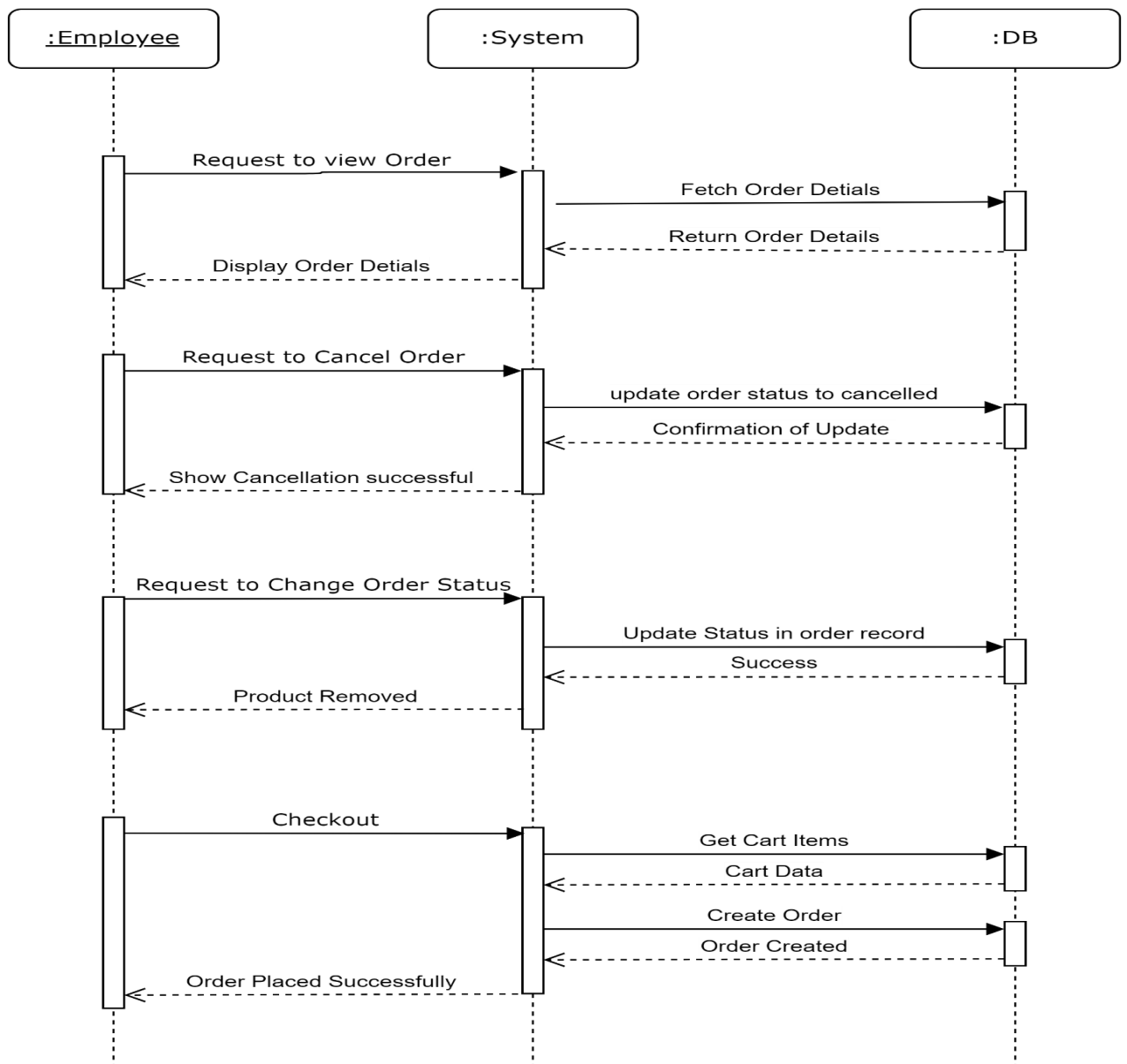


Figure 5: Sequence Diagram for Order Management

- Employee **views order details** (fetched from DB).
- Employee **cancels an order**; system updates status to "cancelled".
- Employee **removes a product**; system updates order record.
- Employee **places an order**; system gets cart data and creates the order.
- All actions involve **system-DB interaction** and return success confirmation.

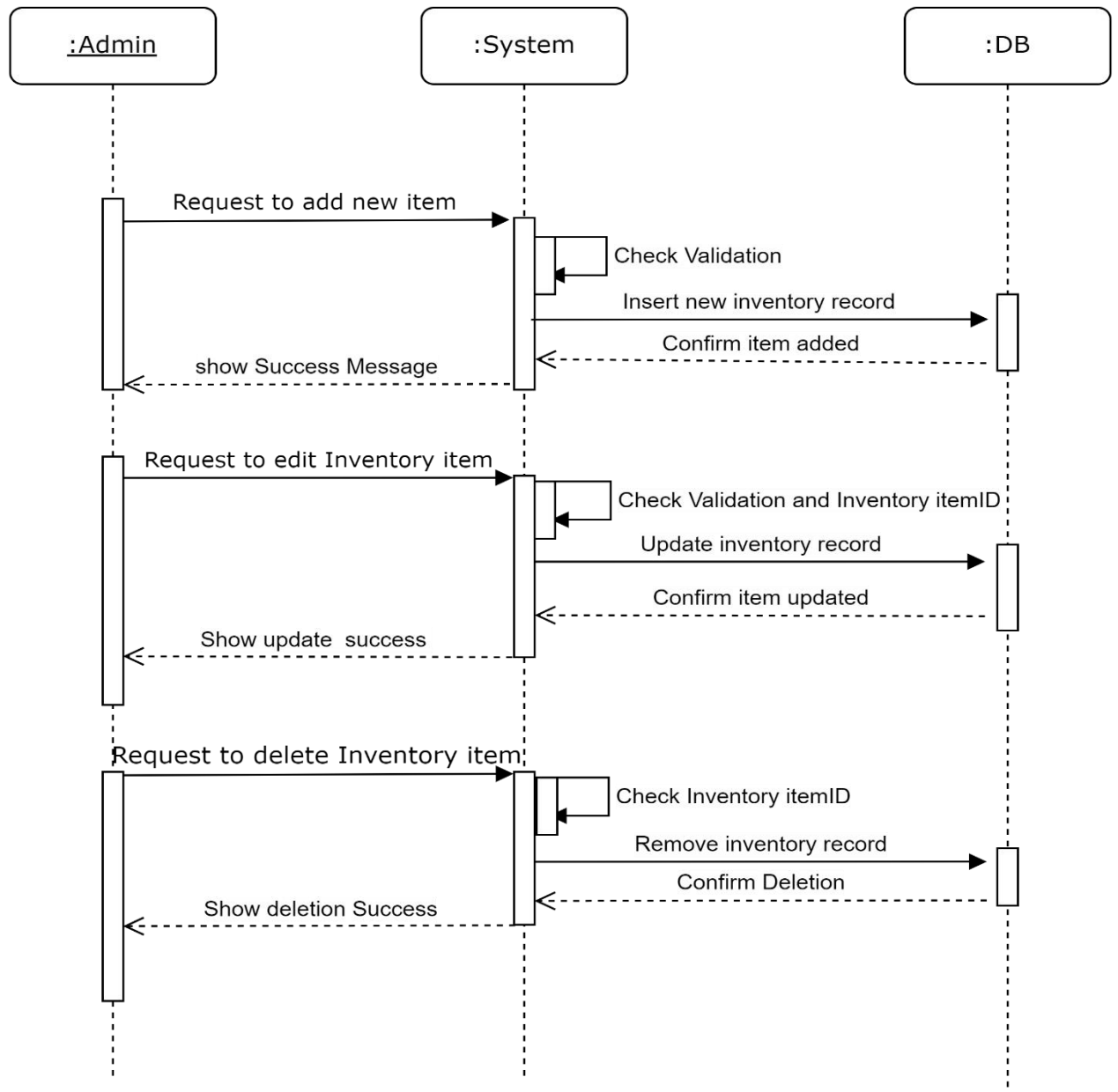


Figure 6: Sequence Diagram for Inventory Management

1. User Management:

The admin can create, edit, and delete users. For each action, the system validates the input, updates the database, and shows a success message.

2. Role Management:

The admin can add and edit roles. The system validates the input, updates the database, and confirms success to the admin.

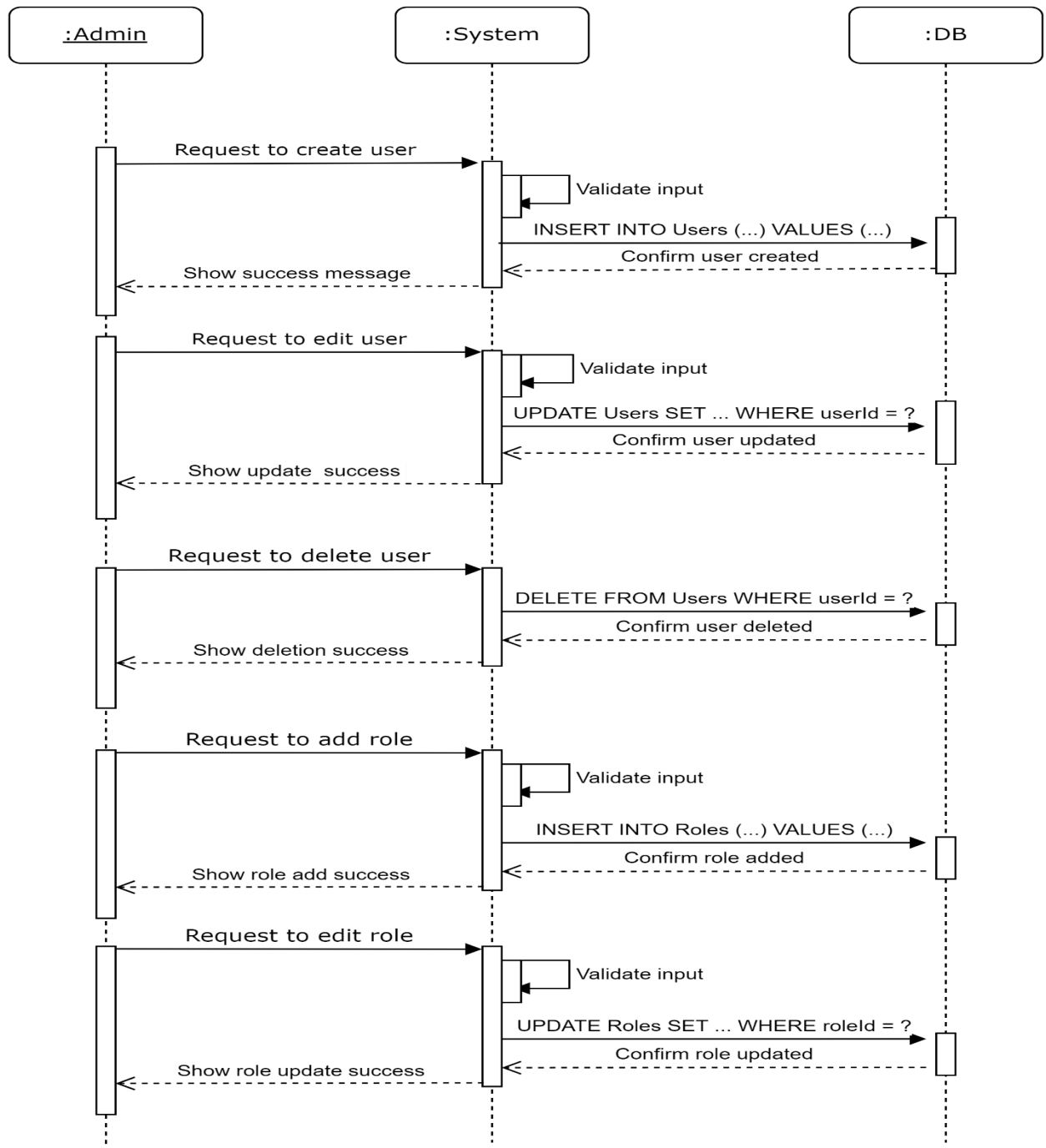


Figure 7: Sequence Diagram for User Role and Management

- The admin can create, edit, and delete users, as well as add and edit roles.
- For each action, the system first validates the input.
- If valid, the system updates the database (insert, update, or delete).
- After the database confirms the change, the system notifies the admin of the successful operation.

Activity Diagram

The activity diagrams capture the dynamic workflows of key processes within the Clean Serve system, such as placing an online order, booking a table, or initiating hygiene checks. They visually represent the sequence of actions, decision points, and parallel activities involved in each process, highlighting the flow from start to finish. These diagrams help clarify the business logic by showing validations, conditional branches, and exception handling steps that occur during execution. By mapping out these workflows, activity diagrams enable better understanding of user interactions and system responses, ensuring that all necessary steps are accounted for and that potential errors or special cases are properly managed. This makes them essential for both designing process logic and guiding the implementation of robust, user-friendly features.

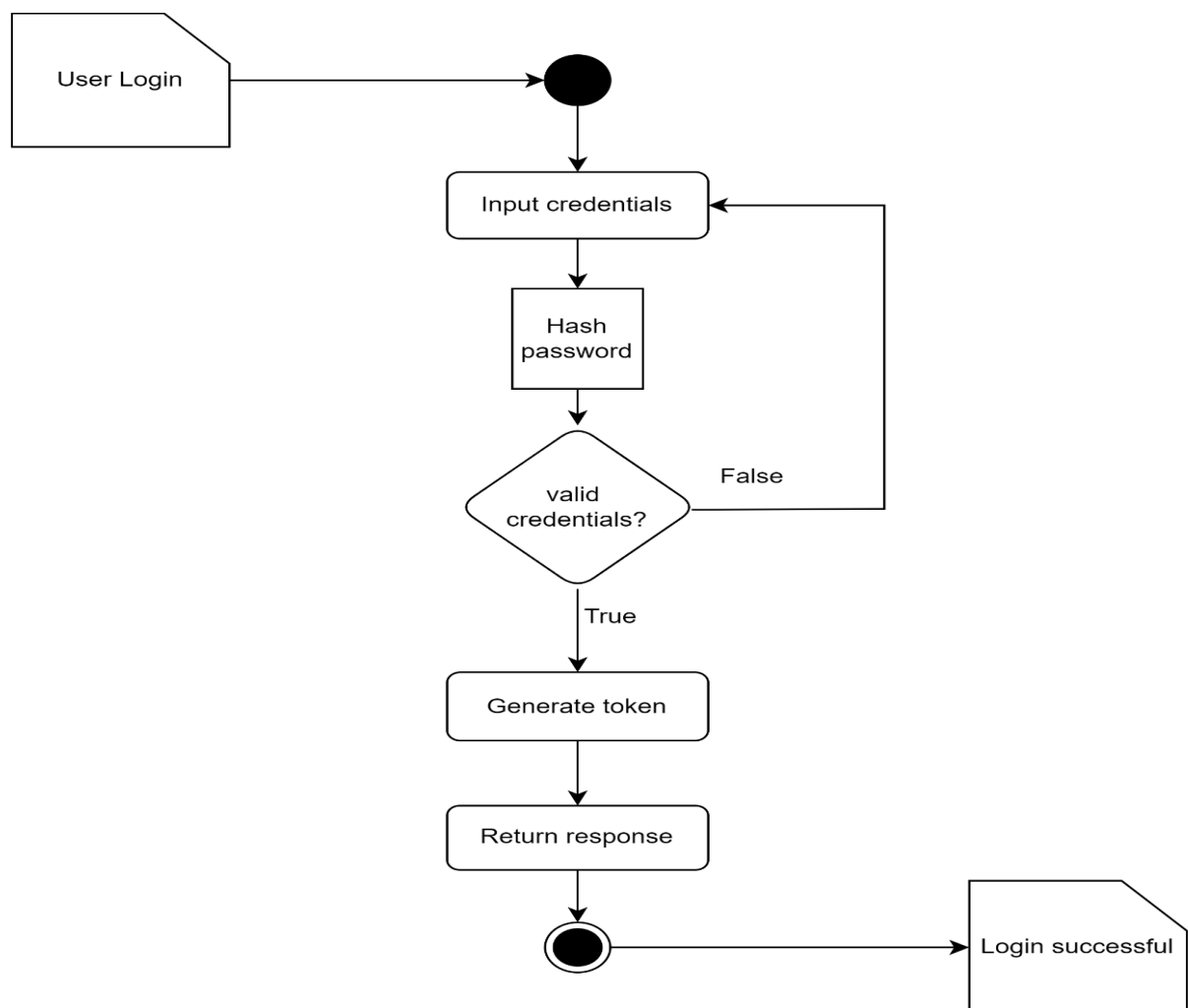


Figure 8: Activity diagram for Login

- User enters credentials, which are hashed and validated.
- If credentials are valid, a token is generated.
- System returns a response, completing the login process.

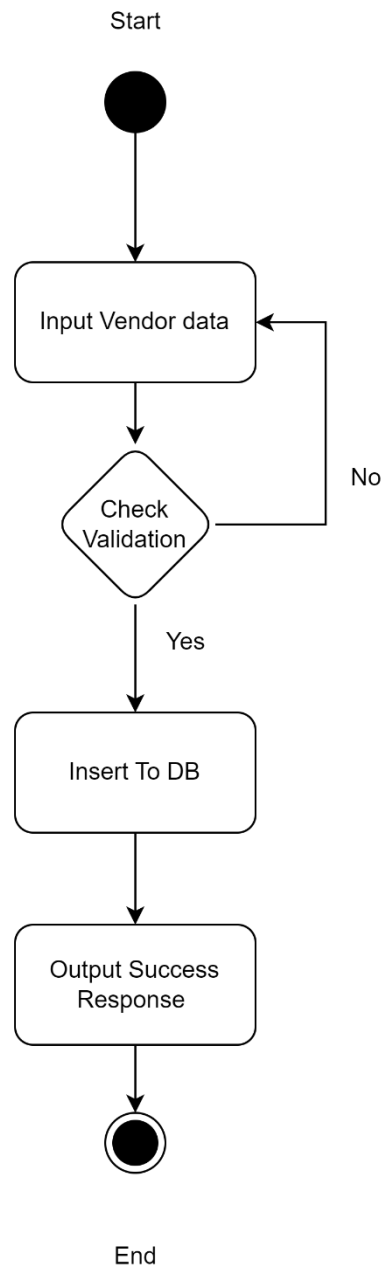


Figure 9: Activity diagram for Add Vendor

- Vendor data is entered and validated.
- If validation passes, the data is inserted into the database.
- A success response is provided after insertion.

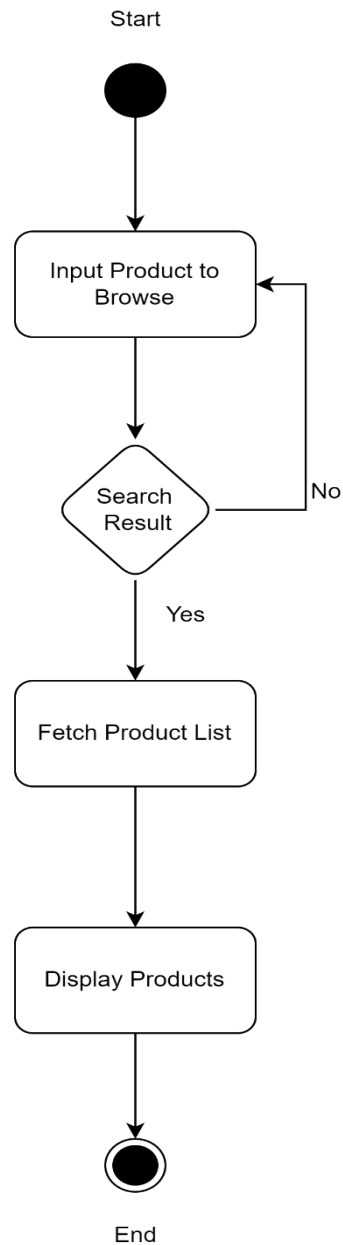


Figure 10: Activity diagram for Browse Products from web

- The user inputs the product they want to browse to initiate the search process.
- The system checks for search results and prompts the user to re-enter if no results are found.
- Upon finding valid results, the system retrieves and displays the product list to the user.

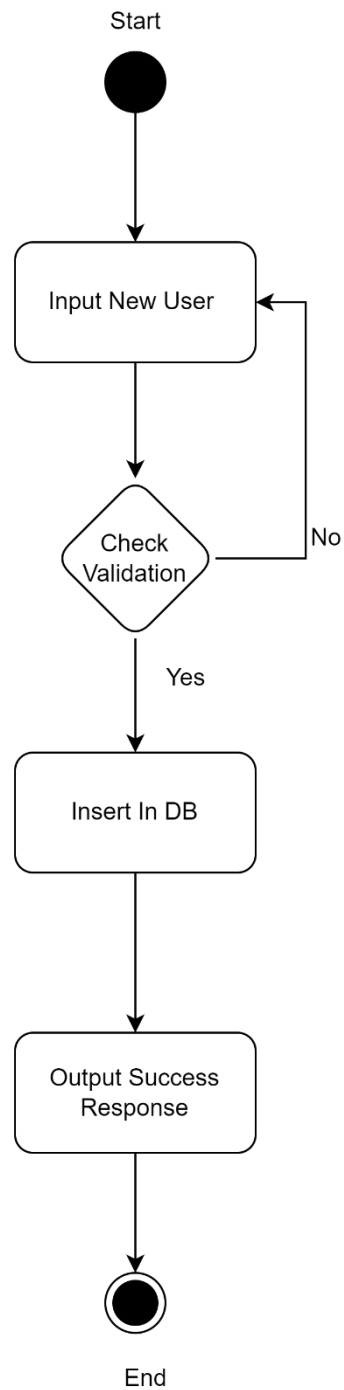


Figure 11: Activity diagram for Add New User

- A new user is entered and their data is validated.
- If validation passes, the user data is inserted into the database.
- After insertion, a success response is provided to complete the process.

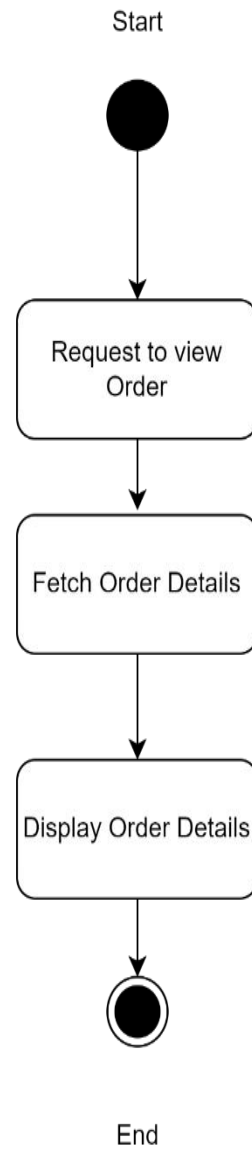


Figure 12: Activity diagram for Order Management

- The user sends a request to view an order.
- The system fetches the order details.
- The order details are displayed to the user.

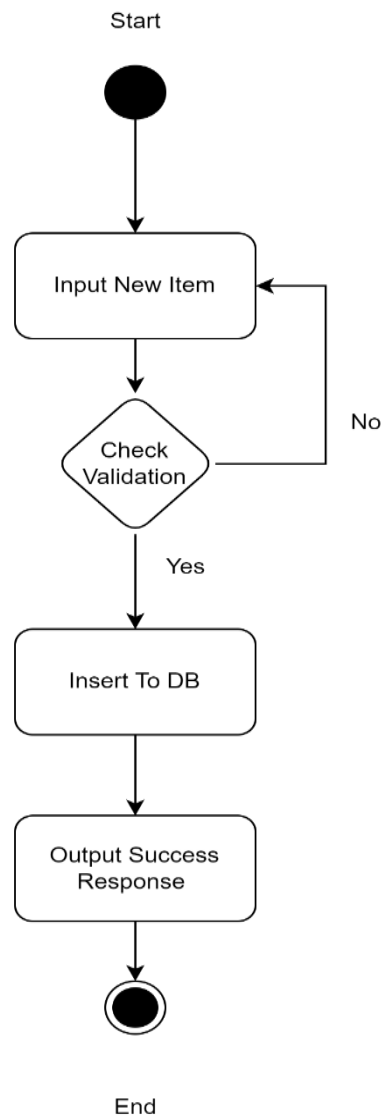


Figure 13: Activity diagram for Add New Product

- User inputs a new item into the system.
- System performs validation on the input.
- If valid, the item is inserted into the database and a success response is returned.

3. Class Diagram

This class diagram illustrates the core structure of the Clean Serve system, highlighting essential entities such as User, Order, Product, Inventory, and HygieneMonitoring. It defines key attributes and methods while showcasing how classes interact through associations, aggregations, and compositions.

The diagram supports functionalities like order processing, stock management, hygiene compliance, and user notifications, forming a clear blueprint for implementing a scalable and maintainable system architecture.

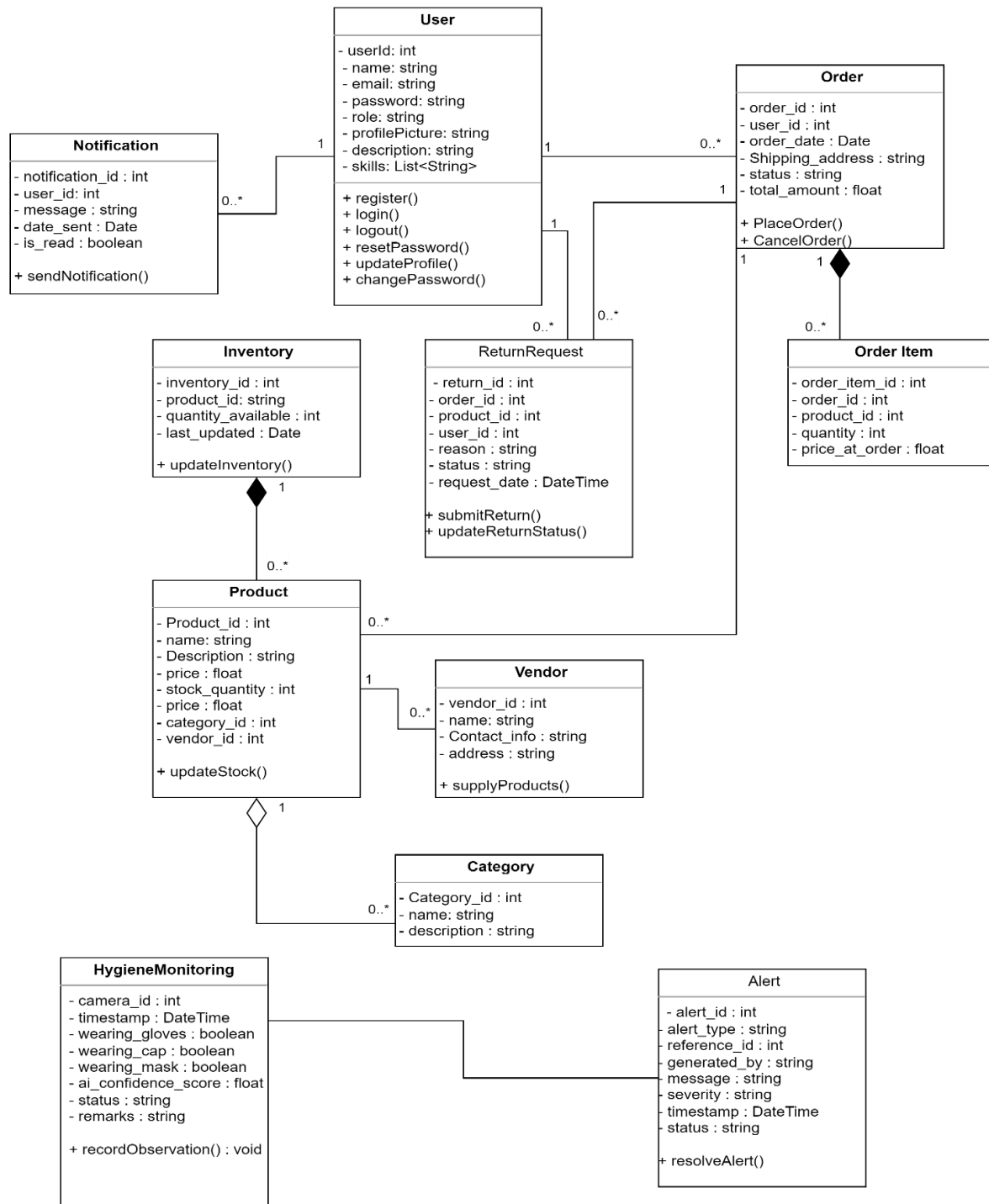


Fig 14: Class Diagram